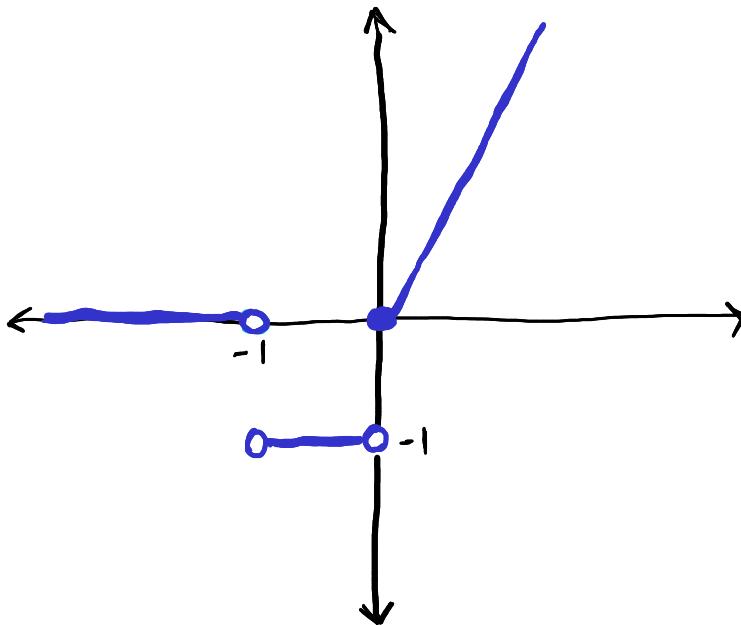
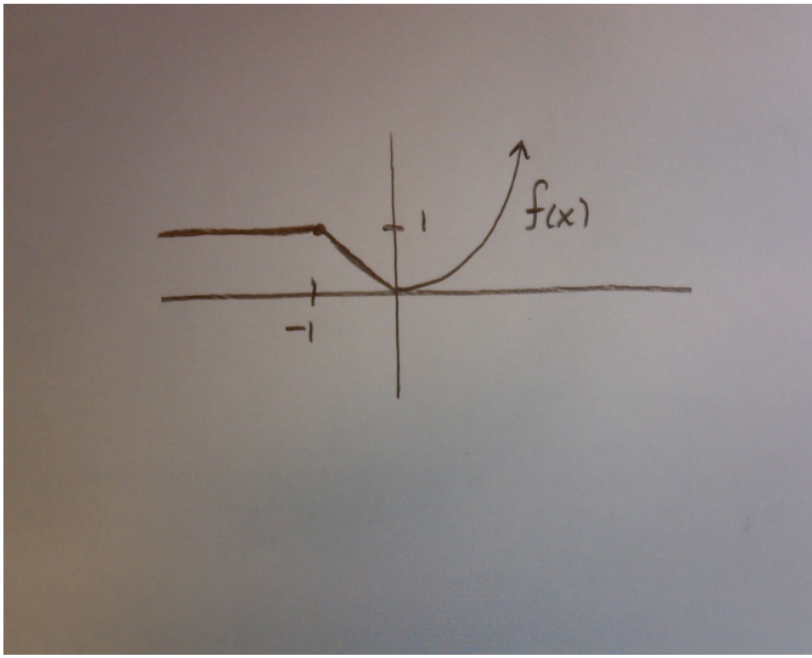


16) Given the graph of $f(x)$ below, sketch the graph of $f'(x)$.



17) The table below gives the percentage of Americans that are under the age 18 along with the years from 1950 to 2000.

t	$P(t)$
1950	31.1
1960	35.7
1970	34.0
1980	28.0
1990	25.7
2000	25.7

a) What is the meaning of $P'(t)$? What are the units?

b) Construct a table with estimates for $P'(t)$.

c) Graph both $P(t)$ and $P'(t)$.

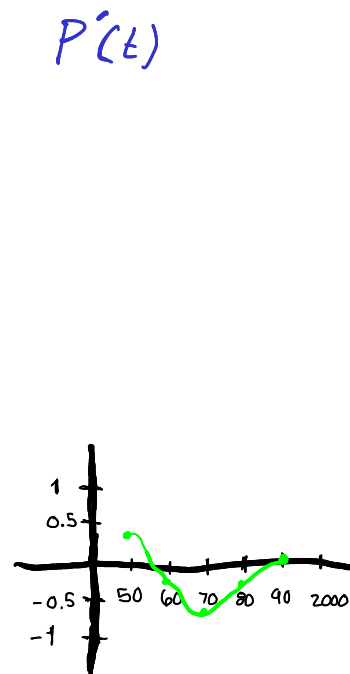
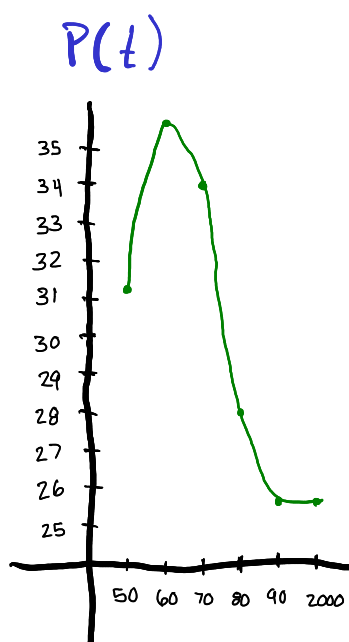
a) How fast the percentage of Americans under age 18 is increasing or decreasing at year t .

Units: percent % $\rightarrow$ percent
year

b)

Year	$P'(t)$
1950	0.46
1960	-0.17
1970	-0.6
1980	-0.23
1990	0

c)



18) Use the definition to find $f'(a)$ for $f(x) = \sqrt[3]{x}$. Then explain why $f'(0)$ does not exist.

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$\lim_{x \rightarrow a} \frac{(\sqrt{x} - \sqrt{a})}{x - a} \cdot \frac{(\sqrt{x} + \sqrt{a})}{(\sqrt{x} + \sqrt{a})}$$

$$\lim_{x \rightarrow a} \frac{\cancel{(x - a)}}{\cancel{(x - a)} (\sqrt{x} + \sqrt{a})}$$

$$\lim_{x \rightarrow a} \frac{1}{\sqrt{x} + \sqrt{a}}$$

$$\lim_{x \rightarrow a} \frac{1}{\sqrt{a} + \sqrt{a}}$$

$$\lim_{x \rightarrow a} \frac{1}{2\sqrt{a}}$$

DNE
(division by zero)

$$f'(0) = \frac{1}{2\sqrt{0}}$$

19) Evaluate $\lim_{x \rightarrow 3} \frac{\sqrt{x+6} - x}{x^3 - 3x^2}$.

I see square root
 $\hookrightarrow$ Mult by Conjugate

$$\lim_{x \rightarrow 3} \frac{(\sqrt{x+6} - x)}{x^3 - 3x^2} \times \frac{(\sqrt{x+6} + x)}{(\sqrt{x+6} + x)}$$

$$\lim_{x \rightarrow 3} \frac{x+6 - x^2}{x^2(x-3)(\sqrt{x+6} + x)}$$

$$\lim_{x \rightarrow 3} \frac{-(x^2 - x - 6)}{x^2(x-3)(\sqrt{x+6} + x)}$$

$$\lim_{x \rightarrow 3} \frac{-(x-3)(x+2)}{x^2(x-3)(\sqrt{x+6} + x)}$$

$$\lim_{x \rightarrow 3} \frac{-(x+2)}{x^2(\sqrt{x+6} + x)}$$

$$\lim_{x \rightarrow 3} \frac{-(3+2)}{3^2(\sqrt{3+6} + 3)}$$

$$\lim_{x \rightarrow 3} \boxed{\frac{-5}{54}}$$

20) Use the definition to find $f'(3)$ for $f(x) = \frac{1}{x+5}$.

$$\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$$

$$\frac{\left(\frac{1}{x+5} - \frac{1}{8} \right)}{(x-3)} \cdot \frac{8(x+5)}{8(x+5)}$$

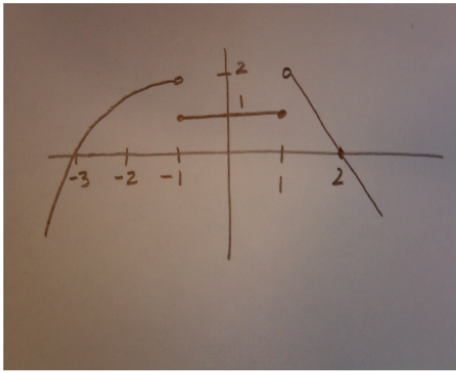
$$\frac{8 - x - 5}{(x-3) \cdot 8(x+5)}$$

$$\frac{\cancel{3-x-1}}{(\cancel{x-3}) \cdot 8(x+5)}$$

$$\frac{-1}{8(x+5)}$$

$$\lim_{x \rightarrow 3} \frac{-1}{8(3+5)} = \boxed{-\frac{1}{64}}$$

21) Given the graph of $f(x)$ below, answer a-d.



a) $\lim_{x \rightarrow -1} f(x) = \text{DNE}$ b) $\lim_{x \rightarrow 1^-} f(x) = 1$ c) $f'(0) = 0$ d) $f'(-1) = \text{DNE}$

22) Given $f(x) = x^3 - 8x^2 + 2x + 3$, find all values of x where $f'(x) = 0$. Also, graph the function.

Power Rule;

$$f'(x) = 3x^2 - 16x + 2$$

to find zero set derivative equal zero
and solve for x

$$3x^2 - 16x + 2 = 0$$

$$x = \frac{-(-16) \pm \sqrt{(-16)^2 - 4(3)(2)}}{2(3)}$$

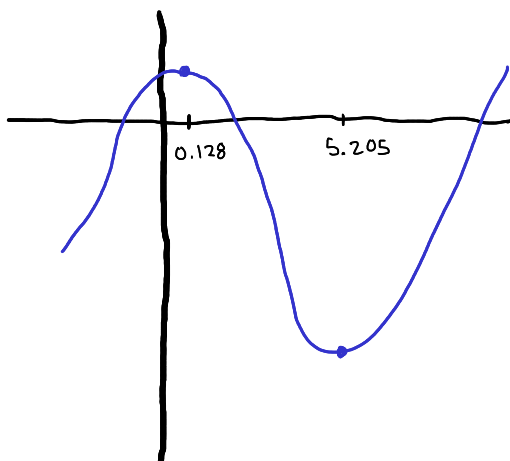
$$= \frac{(16 \pm \sqrt{232})}{6}$$



$$x = 5.205$$



$$x = 0.128$$



23) Find the first derivative of $y = \sqrt[3]{x}(x^2 - x)$.

Rewrite

$$y = x^{1/3} (x^2 - x)$$

$$y = x^{7/3} - x^{4/3}$$

power rule

$$y' = \frac{7}{3}x^{4/3} - \frac{4}{3}x^{1/3}$$

24) Find the equation of the line tangent to $f(x) = x^4 + 2x^2 - x$ when $x = 1$.
 $f'(x)$

Power Rule:

$$f'(x) = 4x^3 + 4x - 1$$

Plug in $x = 1$

$$\begin{aligned} f'(1) &= 4(1)^3 + 4(1) - 1 \\ &= 4 + 4 - 1 \\ &= 7 \end{aligned}$$

$$m = 7$$

Equation

$$\begin{aligned} y - 2 &= 7(x - 1) \\ y - 2 &= 7x - 7 \\ y &= 7x - 7 + 2 \end{aligned}$$

$$y = 7x - 5$$

Find the Point

$$f(1) = 1^4 + 2(1) - 1$$

$$y = 2$$

$$\text{point: } (1, 2)$$

25) Find the first derivative of $f(x) = \sqrt{x}e^x$.
product Rule

$$\left. \begin{aligned} \rightarrow \frac{d}{dx}(\sqrt{x}) &= \frac{1}{2\sqrt{x}} \\ \rightarrow \frac{d}{dx}(e^x) &= e^x \end{aligned} \right\} \begin{aligned} f'(x) &= \overset{A'}{\left(\frac{1}{2\sqrt{x}}\right)} \overset{B}{(e^x)} + \overset{A}{(\sqrt{x})} \overset{B'}{(e^x)} \\ &= e^x \left(\frac{1}{2\sqrt{x}} + \sqrt{x} \right) \end{aligned}$$

26) Find the first derivative of $g(x) = \frac{1-x^2}{2x+3}$.

Quotient Rule

$$\frac{A'B - AB'}{B^2}$$

$$g'(x) = \frac{\overset{A'}{-2x} \overset{B}{(2x+3)} - \overset{A}{(1-x^2)} \cdot \overset{B'}{2}}{\underset{B^2}{(2x+3)^2}}$$

$$g'(x) = \frac{-4x^2 - 6x - 2 + 2x^2}{(2x+3)^2}$$

$$g'(x) = \frac{-2x^2 - 6x - 2}{(2x+3)^2}$$

27) Use the product rule to prove the quotient rule.

> We haven't covered the Chain Rule yet but we will use it.

$$y = x^2$$

$$y' = 2x$$

$$y = (B)^2$$

$$y' = 2 \cdot (B) \cdot B'$$

$$\left(\frac{A}{B} \right)'$$

$$= (A \cdot B^{-1})'$$

$$= A' \cdot B^{-1} + A \cdot (-1) B^{-2} \cdot B'$$

Rewrite

$$= \frac{A' \cdot B}{B \cdot B} + \frac{-A \cdot B'}{B^2}$$

$$= \frac{A'B}{B^2} + \frac{-A \cdot B'}{B^2}$$

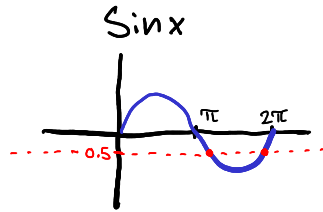
$$= \frac{A'B - AB'}{B^2} \quad \begin{array}{l} \text{Quotient} \\ \text{Rule} \end{array}$$

28) Find a value in $[0, 2\pi)$ so that $\frac{dy}{dx} = 0$ for the function $y = 0.5x - \cos x$.

$$y' = 0.5 + \sin x$$

$$0 = 0.5 + \sin x$$

$$-0.5 = \sin x$$



Calculator

$$\sin^{-1}(-0.5) = -\frac{\pi}{6}$$

(which is the point closest to origin)
↳ on the left

$$\left. \begin{array}{l} \pi + \frac{\pi}{6} = \frac{7\pi}{6} \\ 2\pi - \frac{\pi}{6} = \frac{11\pi}{6} \end{array} \right\} x = \frac{7\pi}{6}, \frac{11\pi}{6}$$

29) Find the equation of the line tangent to $y = x^2 \sin x$ when $x = 2$.

find derivative
1. need a point
2. need Slope

Point

$$y(2) = (2)^2 \sin(2)$$

$$= 3.637$$

$$(2, 3.637)$$

Slope

$$y' = \overset{A'}{2x} \overset{B}{\sin x} + \overset{A}{x^2} \cdot \overset{B'}{\cos x}$$

$$y'(2) = 2(2) \sin(2) + (2)^2 \cos(2)$$

$$m = 1.973 \text{ Slope}$$

$$y - 3.637 = 1.973(x - 2)$$

$$y - 3.637 = 1.973x - 3.945$$

$$y = 1.973x - 0.308$$